Project Documentation

Urdu Fake/Biased News Detector

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1. Introduction

Motivation

Fake and biased news in Urdu spreads fast on WhatsApp, Facebook, and Twitter — harming Pakistan's society, politics, and public health. Unlike English, where extensive research and datasets exist, Urdu, a top-10 most spoken language worldwide with over 230 million speakers, remains a low-resource language for fake news detection.

Reference: https://www.nature.com/articles/s41598-025-91054-4

Challenges specific to Urdu Fake News Detection:

- **Limited digital resources**: Few publicly available Urdu datasets and NLP tools compared to English. Reference: https://norma.ncirl.ie/8739/
- **Rich morphology**: Urdu words have multiple forms, making traditional bag-of-words approaches ineffective. Reference: https://workshop-proceedings.icwsm.org/abstract.php?id=2025_51
- **Code-switching**: Urdu news often mixes English words (e.g., "COVID-19 وائرس"), confusing standard models.
- **Right-to-left script**: Requires special preprocessing pipelines.
- **Multimodal manipulation**: Fake news often pairs false Urdu text with manipulated or mismatched images.
- **Domain diversity**: Fake news spans politics, healthcare, religion, sports, and showbiz. Reference: https://www.nature.com/articles/s41598-026-36771-0

Applications

- Browser extension for Urdu news sites
- Tool for Pakistani journalists

Problem Statement

"To detect whether a given Urdu news item (text) is fake, biased, or real using Naive Bayes and Neural Networks (ANN/CNN), addressing low-resource language challenges."

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2. Background / Literature Review

Recent works in Urdu fake news detection:

Year	Authors	Method	Dataset Size	Key Finding
2025	Naveed et al.	LLaMA 2 fine-tuned	78,409	Achieved **97.8% accuracy** using LLM fine-tuning on "Hook and Bait Urdu" dataset
2025	Shafiq et al.	Ensemble: ELECTRA + mBERT + XLM-RoBERTa (MV-V)	Public dataset	**91.4% accuracy**, outperforming individual PLMs via majority voting with veto
2024	Lohano	DistilmBERT	Large multi-domain	**89% accuracy** — DistilmBERT matches larger models with fewer resources
2024	Scientific Reports	XLM-RoBERTa + concatenated GloVe	14,178 (15 domains)	**96.2% accuracy** — State-of-the-art for Urdu FND

Gap in existing work addressed by this project:

- Existing datasets are preprocessed — this model uses **raw, self-collected data**.
- None explicitly address **bias detection** (vs. only fake/real binary classification).

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3. Dataset Details

Important note: *Preprocessed datasets available online are not used. Raw data was collected manually from original Urdu news sources.*

Data Collection Strategy

Manually collected **1,000+ Urdu news articles** (text) from:

Source Type	Sources	Count	Label
Real News	Daily Jang (جنگ), Daily Express (روزنامہ ایکسپریس), Geo News (جیو نیوز), ARY News (اے آر وائی), Dunya News (دنیا نیوز), UrduPoint	~400	Real
Fake News	The Balochistan Post Media Network (extremist propaganda), Deepfake videos (e.g., Shehbaz Sharif), Doctored audio recordings (e.g., Kashmir screening claim), Flood reporter fake video with BBC Urdu branding	~350	Fake
Biased News	Nawa-i-Waqt (نوائے وقت), Daily Jang (جنگ), Daily Express (ایکسپریس), Daily Khabrain (خبریں), Daily Pak (پاکستان), Daily Ausaf (اوصاف), Waqt News	~350	Biased

Raw Collection Method

Text: Scraped raw HTML (BeautifulSoup, Selenium) — no cleaning at collection stage.

Data Distribution by Domain

Domain	Count
Politics (سیاست)	~160
Health (صحت)	~120
Religion (مذہب)	~100
Sports (کھیل)	~100
Crime (جرائم)	~80
Technology (ٹیکنالوجی)	~80
Other (Business, Education, Travel, Environment, Science, etc.)	~360

Label Definitions

Label	Urdu Term	Description	Count
Real	**حقیقی**	Factually correct, neutral tone, verifiable sources	~400
Fake	**جعلی**	Demonstrably false claims, fabricated events/quotes	~400

Biased	**متعصبانہ**	True facts but misleading framing, emotionally charged language	~200

Dataset Format

The dataset is stored as a CSV file with the following structure:

text	label
"حکومت نے نیا قانون منظور کر لیا"	Real
"مخالف جماعت نے جھوٹے الزامات لگائے"	Biased
"واٹس ایپ پر افواہ پھیل رہی ہے"	Fake

Example Raw Entry (Fake News)

Urdu Text: "کورونا ویکسین میں چپ لگائے جانے والے مائیکرو چپس ہیں جو حکومت کو آپ کی لوکیشن بتائیں گے"
English Translation: "COVID-19 vaccine contains microchips that will tell the government your location"

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4. Project Structure

```
Urdu_Fake_News_Project/
│
├── preprocess.py       # Step 1: Clean and normalize Urdu text
├── features.py         # Step 2: Build vocabulary, TF-IDF, sequences
├── naive_bayes.py      # Step 3: Multinomial Naive Bayes classifier
├── ann.py              # Step 4: Artificial Neural Network
├── cnn.py              # Step 5: 1D Convolutional Neural Network
├── train_test.py       # Step 6: Run all models, save results
├── compare.py          # Step 7: Compare models, generate charts
│
├── data/
│   ├── raw/            # Put your dataset.csv here
│   └── processed/      # Auto-generated: features, models, results
│
└── utils/
    ├── urdu_normalizer.py  # Urdu character normalization + stopwords
    └── metrics.py          # Accuracy, Precision, Recall, F1 (from scratch)
```

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5. Methodology

Complete Pipeline Overview

```
Raw CSV Dataset
    ↓
preprocess.py → Clean Urdu text (normalize, remove stopwords)
    ↓
features.py → TF-IDF + Word Sequences + Embeddings
    ↓
┌──────────────────────────────────────────┐
│ naive_bayes.py │ ann.py │ cnn.py │
└──────────────────────────────────────────┘
    ↓
train_test.py → Orchestrate full pipeline
    ↓
compare.py → Evaluate & compare all models
```

Step-by-Step Methodology

Phase 1: Text Preprocessing (`preprocess.py`)

Takes raw Urdu news text and prepares it for machine learning.

What happens:

1. Reads the CSV file from `data/raw/`
2. Removes emojis, special symbols, URLs, and HTML code
3. Normalizes Urdu characters (e.g., آ and ا treated as the same letter)
4. Removes stopwords (common words like "ہے", "کا", "میں")
5. Saves cleaned texts and labels to `data/processed/dataset.npz`

Example transformation:

Input: "حکومت نے نیا قانون پاس کیا ہے!!!"
Output: "حکومت نیا قانون پاس کیا"

Phase 2: Feature Extraction (`features.py`)

Converts text into numerical representations that models can process.

What happens:

1. **Vocabulary building** — assigns a unique ID to every word
2. **TF-IDF features** — measures word importance per document (for Naive Bayes)
3. **Padded sequences** — makes all texts the same length (200 words)
4. **Word embedding matrix** — 300-dimensional vectors per word (for ANN and CNN)
5. **Train/Validation/Test split** — 70% / 10% / 20%
6. Saves all features to `data/processed/features.npz`

Phase 3: Naive Bayes Classifier (`naive_bayes.py`)

A fully from-scratch implementation of Multinomial Naive Bayes using NumPy only (no scikit-learn).

Architecture:

- Computes log-prior probabilities: `log P(class)`
- Computes log-likelihood: `log P(word | class)` with Laplace smoothing
- Inference via: `log P(c|x) ∝ log P(c) + x · log P(w|c)`
- Grid search over alpha values: `{0.01, 0.1, 0.5, 1.0, 2.0, 5.0}`
- Final model trained on train + validation combined

Expected accuracy: 60–68%

Phase 4: Artificial Neural Network (`ann.py`)

A feedforward neural network with word embeddings, implemented from scratch.

Architecture:

```
Input Words → Embedding Layer → Dense(128) → Dense(64) → Output(3 classes)
              (300-dim)          + ReLU       + ReLU     (Real/Biased/Fake)
```

Training details:

- Loss: Cross-entropy
- Optimizer: SGD with Momentum
- Dropout: 30% (to prevent overfitting)
- Weight initialization: He initialization

- Early stopping: Stops if validation accuracy doesn't improve for 5 epochs
- Batch size: 16
- Best weights are restored automatically

Expected accuracy: 68–75%

Phase 5: Convolutional Neural Network (`cnn.py`)

A 1D CNN designed to detect local patterns (phrases) in Urdu text.

Architecture:

Input Words → Embedding → Conv1D(128 filters, size=5) → GlobalMaxPool → Dense(64) → Output

Key concept: The Conv1D layer slides a 5-word window across each article, learning to recognize suspicious phrases regardless of their position. 128 filters means the model can simultaneously look for 128 different patterns.

Example learned patterns:

Filter #1: Detects "پھیل" + "افواہ" → strong signal for Fake
Filter #2: Detects "اعلان" + "حکومت" → strong signal for Real
Filter #3: Detects "الزام" + "مخالف" → strong signal for Biased

Total parameters: ~277K
Expected accuracy: 70–78%

Phase 6: Orchestrator (`train_test.py`)

Runs the entire pipeline in sequence:

1. Preprocessing → Feature extraction → Naive Bayes → ANN → CNN
2. Collects and saves all results

Running `train_test.py` alone executes the full pipeline.

Phase 7: Comparison & Evaluation (`compare.py`)

Generates visual comparisons of all three models.

Outputs:

- `comparison_chart.png` — Accuracy + F1 bar chart
- `curve_ann.png` — ANN training curve (loss over epochs)
- `curve_cnn.png` — CNN training curve (loss over epochs)
- Console table with all metrics side-by-side

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6. Urdu NLP Utilities

`utils/urdu_normalizer.py`

Handles Urdu script normalization:

Problem	Fix
آ/أ/إ → ا	All Alef forms → standard Alef
ي/ئ/ى → ی	Arabic Yeh → Urdu Yeh
ك → ک	Arabic Kaf → Urdu Kaf
Latin digits 123 → ۱۲۳	Latin to Urdu numerals
Diacritics removal	Optional zabar/zer/pesh removal
Stopwords (75+)	"ہے", "کا", "میں", "اور", etc.

`utils/metrics.py`

All evaluation metrics implemented from scratch (no scikit-learn):

- **Accuracy**: `correct_predictions / total_predictions`
- **Precision**: For each class: `TP / (TP + FP)`
- **Recall**: For each class: `TP / (TP + FN)`
- **F1 Score**: `2 × (Precision × Recall) / (Precision + Recall)`
- **Confusion Matrix**: Tabular breakdown of class-level predictions

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7. Expected Results

Model	Accuracy	Notes
Naive Bayes	60–68%	Simple, fast, assumes word independence
ANN	68–75%	Learns word relationships, better than NB
CNN	70–78%	Best at detecting patterns anywhere in text

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8. How to Run

Step-by-step:

python preprocess.py
python features.py
python naive_bayes.py
python ann.py
python cnn.py
python train_test.py
python compare.py

Or all at once:

python train_test.py

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9. References

1. Scientific Reports (2026). *Verifying Urdu news authenticity using deep learning with concatenated BERT and GloVe embedding*. Nature. https://www.nature.com/articles/s41598-026-36771-0

2. Naveed, A. et al. (2025). Benchmarking Hook and Bait Urdu news dataset for domain-agnostic multilingual fake news detection using large language models. Scientific Reports. https://www.nature.com/articles/s41598-025-91054-4

3. Shafiq et al. (2025). Multi-domain Urdu fake news detection using pre-trained ensemble model. Scientific Reports. https://www.nature.com/articles/s41598-025-91054-4

4. Lohano, J. (2024). Automated Detection of Fake News in Urdu Language Using Pre-Trained Transformer Models. Masters thesis, National College of Ireland. https://norma.ncirl.ie/8739/

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